MAX HEAP SORT:

Max Heap Sort is a comparison-based sorting algorithm that first creates a Max Heap, where the largest element is at the root. The largest element is then repeatedly moved to the end of the array

ALGORITHM:

1. Start with the given array.

2.Build a Max Heap from the array.

3.The largest element will be at the root (arr[0]).

4.Swap the root with the last element.

5.Reduce the heap size by one.

6.Apply Heapify to restore the Max Heap.

7.Repeat until all elements are sorted.

Time Complexity:

Case	Complexity
Best	O(n log n)
Average	O(n log n)
Worst	O(n log n)

Space Complexity:

Space Complexity **= O(1)**

CONCLUSION:

Max Heap Sort provides a guaranteed O(n log n) running time and does not require extra array space. It is useful when consistent performance and low auxiliary memory usage are important.

OUTPUT:

```
Enter number of elements: 6
Enter elements:
50
30
40
20
60
10
Sorted array: 10 20 30 40 50 60

...Program finished with exit code 0
Press ENTER to exit console.
```